

Mohammad Ummair

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Research Interests

ML Systems and Large-Scale Distributed Training/Inference Infrastructure – specifically: efficient model parallelism (intra- and inter-operator), communication optimization for cross-device tensor computation, LLM serving throughput and scheduling, and load-balancing for long-context training. Motivated by systems like Alpa, vLLM, and DistServe; seeking to build infrastructure that makes training and deploying large models faster, cheaper, and more scalable.

Education

- **Indian Institute of Technology (IIT), Bombay** Mumbai, India
Master of Technology, Computer Science & Engineering; CPI: 8.74/10 Jul. 2021 – Jul. 2023
 - **Thesis:** Enhancing Robustness of Smart Contracts Through Declarations (Advisor: Prof. R.K. Shyamasundar)
 - **Admission:** Admitted via GATE CS 2021 – All India Rank **95** out of 101,922 candidates (Top 0.1%)
 - **Relevant Coursework:** Advanced Blockchain Technology, Foundations of Machine Learning, Algorithms & Complexity, Design & Engineering of Computing Systems, Introduction to Blockchains & Cryptocurrencies
- **Government College of Engineering, Aurangabad** Aurangabad, India
Bachelor of Engineering, Information Technology; CPI: 7.5/10 Jul. 2017 – Jul. 2021
 - **Achievement:** Winner, Smart India Hackathon 2020 (Process Modelling, GAIL India Pvt. Ltd.)

Publications

- **Enhancing Robustness of Smart Contracts Through Declarations** [Springer, 2024]
R.K. Shyamasundar, Mohammad Ummair, S. Borse
 - **Declarative Security Framework:** Proposed a novel Solidity language extension (Solidity+) that abstracts run-time monitoring via high-level declarations (NONRENT, PARALLEL, ACCESS, INV), enabling automatic instrumentation of security logic at compile time without altering programmer-visible contract structure.
 - **Transformation Pipeline:** Designed and implemented a source-to-source preprocessor that semantics-preservingly transforms Solidity+ programs into hardened Solidity, embedding assertions as an on-chain *proof skeleton* – realizing a proof-carrying code paradigm for immutable blockchain contracts and enabling source-level debugging rather than EVM bytecode inspection.
 - **Vulnerability Taxonomy & Algorithmic Mitigations:** Formally characterized six Solidity vulnerability classes – reentrancy, arithmetic overflow/underflow, unchecked send, tx.origin misuse, block.timestamp dependence, and concurrency races – and developed targeted transformation algorithms for each, covering critical failure modes responsible for the \$60M DAO hack and \$30M Parity wallet exploit.
 - **Concurrency & Access-Order Specification:** Introduced PARALLEL and NONRENT declaration clauses to formally specify and enforce concurrency constraints at the language-semantics level, enabling coarse-grained concurrent invocation policies and non-reentrant guarantees without manual mutex instrumentation by the programmer.
 - **Comparative Evaluation Against Static Analyzers:** Empirically evaluated Solidity+ against leading static analysis tools (Mythril, Oyente, Manticore, Porosity) on real-world smart contract datasets, demonstrating that a declaration-driven run-time enforcement approach addresses vulnerability classes that purely static methods cannot soundly detect.

Industry Research & Engineering Experience

- **Apple Inc.** Bangalore, India
Software Engineer – Systems & Infrastructure Jul. 2023 – Present
 - **Distributed Task Execution Framework:** Designed and implemented a multi-node parallel execution system for a silicon firmware validation pipeline used by 100+ engineers across Apple's Silicon Engineering Group (SEG), including the GPU and CPU silicon teams. Engineered task scheduling, inter-node communication, and load distribution logic, achieving a **30% reduction** in end-to-end latency – directly analogous to worker-level scheduling in distributed ML training systems.
 - **LLM Inference Pipeline Engineering:** Designed and deployed an end-to-end LLM inference serving system on Apple Foundation models, engineering in-memory batching, query routing, and retrieval-augmented generation pipelines achieving **30% higher accuracy** and **40% faster query resolution** – gaining hands-on insight into LLM serving bottlenecks and throughput optimization.
 - **Queue-Based Scheduling & Latency Optimization:** Built a message-queue-driven task dispatch system (RabbitMQ) with incremental execution logic that reduced average task wait time from **100s to 10s** by eliminating redundant work – a problem structurally similar to straggler mitigation in pipeline-parallel training.
 - **MCP Server for Distributed Framework Access:** Designed and built a Model Context Protocol (MCP) server exposing internal distributed execution framework APIs to LLM-based agents, enabling silicon engineers to interact with and orchestrate validation tasks through natural language – bridging LLM tool-use with production infrastructure.
 - **Infrastructure Observability:** Developed real-time telemetry and structured logging in Go and Python across distributed node infrastructure, sustaining **99.9% system reliability** in production.
 - **GPU/CPU Architecture Exposure:** Through direct, day-to-day collaboration with Apple's GPU and CPU silicon teams on their internal tooling, developed working knowledge of GPU/CPU microarchitecture and the HW/SW co-design trade-offs that shape silicon bring-up.

- **Tata Consultancy Services** India
Project Intern *Feb. 2021 – May 2021*
 - **Time Series Forecasting:** Deployed hierarchical time series models to forecast client export demands, enabling proactive inventory management and improving demand accuracy for daily operations.
- **TechInvento** India
Machine Learning Intern *May 2019 – Jun. 2019*
 - **Computer Vision for Document Digitization:** Built an automated CV pipeline to digitize handwritten documents, saving the client an estimated **\$1.5M/month**; deployed as a Node.js web application for production monitoring.

Research Projects

- **High-Throughput LSM-Tree Storage Engine [GitHub]:** Engineered a persistent Log-Structured Merge-tree optimized for write-heavy workloads with Bloom filter-based I/O elimination and tiered compaction – building intuition for storage-compute trade-offs central to ML checkpoint and activation management.
- **Single-Machine Linearizable Key-Value Store [GitHub]:** Built a fault-tolerant KV store guaranteeing **linearizable consistency** and **exactly-once semantics** under failures via write-ahead logging and operation identifiers – grounding understanding of consistency models relevant to distributed parameter servers.
- **Distributed Ledger & Consensus Simulation:** Simulated a multi-node P2P network with Proof-of-Work consensus and transaction propagation; benchmarked throughput, latency, and resilience under adversarial conditions, developing hands-on experience with distributed coordination protocols.
- **Multi-Threaded KV Server over gRPC (Course: CS744):** Implemented a concurrent client-server KV store with pluggable caching strategies and a load generator to study throughput/latency under increasing concurrency – mirroring performance analysis methodology used in ML systems evaluation.
- **Asset Performance Monitoring with ML (Smart India Hackathon 2020 – Winner):** Developed a real-time APM application integrating ML to optimize mechanical device performance for industrial use; selected as winning solution for GAIL India Pvt. Ltd.'s process modelling problem statement.

Teaching & Academic Service

- **IIT Bombay – Department of Computer Science** Mumbai, India
Teaching Assistant *2021 – 2023*
 - **CS 765: Introduction to Blockchains, Cryptocurrencies & Smart Contracts (Prof. Vinay Ribeiro):** Conducted lab sessions, graded assignments and exams, and provided mentorship and guidance to graduate students on blockchain concepts and smart contract development.
 - **CS 744: Design & Engineering of Computing Systems (Prof. Mythili Vutukuru):** Administered and graded lab assignments, conducted vivas (oral examinations), evaluated student answers, and mentored students on systems design principles and debugging.
 - **CS 251: Software Systems (Prof. Amitabha Sanyal):** Assisted with lab sessions, grading, and student mentorship.
 - **CS 254: Digital Logic Design Lab (Prof. Virendra Singh):** Supported practical lab instruction and student evaluations.
- **Unacademy** India
Prime Educator – GATE CS Preparation *May 2021 – Jun. 2021*
 - **Mentorship:** Mentored a cohort of 20 students for GATE CS; designed Customized Daily Practice Problems and performance improvement plans for underperforming students.
- **IIT Bombay – Department of Computer Science** Mumbai, India
Department Placement Coordinator & Web Secretary *Jul. 2022 – Jul. 2023*
 - **Placement Coordination:** Managed end-to-end campus placement logistics for the Computer Science department ensuring smooth execution across 100+ recruiting companies.
 - **Web Secretary:** Maintained and updated official Computer Science department web presence.

Technical Skills

- **Languages:** Python, C++, Go, JavaScript/TypeScript, SQL, Java, C
- **Frameworks & Libraries:** FastAPI, Flask, React, Node.js, Haystack, LangChain, Hugging Face, gRPC
- **Systems & Infrastructure:** AWS (EC2, S3), Docker, Kubernetes, PostgreSQL, MongoDB, RabbitMQ, CI/CD, Git
- **Research Tools:** L^AT_EX, MATLAB (basic), Solidity, EVM tooling